Group Project Brief: one method, one question, one decision

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Course: New Media Technology and Analytics (20001920076) ·**Fall 2026** ·Tongji School of Economics and Management (SEM) ·**Weight:** 30% of the course grade ·**Teams:** up to 7 students, about 12 teams *Companion documents (same folder):* `project_question_menu.md`, `project_rubric.md`, `project_report_template.md`, `peer_evaluation_form.md`, `project_data_dictionary.md`, `project_data_guidebook.md`. *The artificial intelligence (AI) use policy (*`ai_use_policy.pdf`*, distributed with the Week 1 pack) applies to everything you submit.*

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1. What you are doing

Each team (up to 7 students, formed in Week 1) picks **one new-media marketing question** and answers it with **one method family from Weeks 3-8**:

- Week 3: Association and Rules
- Week 4: Classification
- Week 5: Regression
- Week 6: Clustering
- Week 7: Text Analysis and Natural Language Processing (NLP)
- Week 8: Social Network Analysis

Four deliverables: a **pitch** (8 min + 3 min of questions and answers (Q&A), Weeks 10-11), a **written report** (8-12 pages), a **notebook** (Python or Stata, fully reproducible), and a confidential **peer evaluation** from each member.

Three questions decide whether a project succeeds: *What is the question worth? What does the data say? What should someone do differently on Monday morning?*

2. Timeline

#	Milestone
1	brief out Week 2 (Thu 2026-09-24)

#	Milestone
2	questions at the next session
3	topic claim by email top-2 picks, deadline Wed 2026-10-14 23:59
4	confirmed topics posted Week 5 (Thu 2026-10-15)
5	milestone check-in Week 6 (Thu 2026-10-22, in class)
6	slides due before class Week 10 (Thu 2026-11-19)
7	presentations: Week 10 (8 teams) and Week 11 (Thu 2026-11-26, 4 teams), 8 min pitch + 3 min Q&A
8	report + peer evaluation due Thu 2026-11-26 23:59

Presentation order within each session is drawn in class.

3. The dataset and question menu

The course project dataset is the **Lumen Beauty full Sep-Nov 2026 operations export**: the Douyin skincare brand from the Week 2 case, extended from the September teaching slices to the store's complete first quarter. Five tables:

- `lb_streams`: one row per livestream session. 78 sessions, with viewers, comments, shares, and gross merchandise value (GMV) per session.
- `lb_order_lines`: one row per stock keeping unit (SKU) line in an order. 4,920 lines across the livestream and store channels.
- `lb_users`: one row per registered user. 1,000 users with arrival channel, age band, city tier, and spend summary.
- `lb_comments`: one row per comment posted during a stream. 10,866 comments with the platform's sentiment labels.
- `lb_follows`: one row per follow or share event. 3,020 edges in the store's social graph.

The five comma-separated values (CSV) files live in the `data/` folder of the project kit: `lb_streams.csv`, `lb_order_lines.csv`, `lb_users.csv`, `lb_comments.csv`, `lb_follows.csv`. Column-level documentation lives in `project_data_dictionary.md`. The question menu `project_question_menu.md` lists sixteen ready-made questions (Q1-Q16) across the six method families, plus Q17, your own question. If your question needs data beyond these five tables, read `project_data_guidebook.md` before you claim a topic: it lists public datasets whose licenses were checked for this course, and the one-page data plan that every such team sends with its claim.

4. Deliverables in detail

Pitch. 8 minutes of pitch + 3 minutes of Q&A, in Week 10 (8 teams) or Week 11 (4 teams). Slides are due before class Week 10 (Thu 2026-11-19). Upload them to Canvas before class on the due date. Show a result, not just slides: a number, a segment profile, a network map. The Q&A can put any question to any member. Everyone on the team is accountable for the entire analysis.

Report. 8-12 pages, executive-readable, seven sections: problem & what it is worth ·data ·method & assumptions ·findings ·recommendation ·ethics & compliance implications ·AI-use appendix. Write it for a manager, not a programmer. The notebook carries the code. Build it on `project_report_template.md`. Submit it as a PDF via Canvas.

Notebook. Python or Stata, your team's choice. A Python notebook must pass *Kernel > Restart & Run All* with no errors; a Stata do-file must run top to bottom cleanly. Every number in the report and on the slides must reproduce from it.

Peer evaluation. Each member submits their own confidential form (`peer_evaluation_form.md`) as a separate file in the same Canvas submission as the report. It individualizes the group score (§5).

5. Grading

Grading: group score (100) = report 40 + pitch 30 + technical 20 + investor vote 10; individual = group score x peer factor (avg teammate rating / team average, capped 0.80-1.20, result capped at 100); no form => own factor <= 1.0

The instructor scores the report, the pitch, and the technical work through seven lenses (**business relevance, methodological correctness, theory framing, actionability, communication, AI-use craft, ethics & compliance**) with level descriptors in `project_rubric.md`. The four components match what Week 2 announced: business relevance lives in the report, methodological correctness feeds both the report (40) and the technical work (20), communication lives in the pitch, and teamwork in the peer factor.

6. Rules

AI use. Governed by `ai_use_policy.pdf`: AI assistance is encouraged for coding, debugging, understanding, and drafting; anything material is disclosed in the report's AI-use appendix; nothing is submitted that the team cannot explain live; platform personal data never enters an external AI service.

Academic integrity. Between teams: discuss concepts, never share artifacts: no shared notebooks, data files, slides, or text. Cite what you borrow. Collusive investor voting voids the votes.

Data. Dataset: course project dataset (Lumen Beauty full Sep-Nov operations export, five tables) or your own proposal; public, properly-licensed, or realistic synthetic data only; no scraped personal data. Any data other than the course project dataset needs an approved data plan: see `project_data_guidebook.md`.

7. Advice from the course design

- **Start from the question, not the method.** "Livestreaming marketing" is a topic; "do mid-tier streamers outsell top-tier streamers per viewer for skincare?" is a question. Push your team from the first toward the second.
- Choose the question where you can already see the Monday-morning recommendation taking shape.
- Put a number on the stakes early: "this decision is worth ¥X per month" sharpens everything that follows.
- Go deep, not wide: a single method applied rigorously and defended clearly will outscore a shallow tour of several. Marks follow how well you execute and justify, not how many techniques you list.
- State your assumptions before your results: a sensible support threshold, a fair comparison group, a defensible k.
- Use the Week 6 milestone check-in: arrive with the data loaded and one result in hand, leave with a plan.
- Drill the Q&A as much as the pitch itself. Investors make up their minds while you answer, not while you present.
- Divide the work so that every member genuinely owns a part. The peer evaluation rewards it.

Questions: email `huwei72@tongji.edu.cn`, cc `2510096@tongji.edu.cn` (the teaching assistant), with the subject line `NMTA: project <your topic>`.